Arthur Mourgue

Robotics Software Engineer | Perception and Navigation | 6-month internship from April 2027
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Education

Sorbonne Université **2026 to 2027**
MSc in Robotics *Paris, France*

- Research project on world models (JEPA) for robot swarms at ISIR.

INSA Lyon **2023 to 2026**
MSc in Electrical Engineering, specialization in Artificial Intelligence *Lyon, France*

- Built the BioMAÉ vision pipeline, which detects gammarids in water with 90% accuracy (GitHub).

Université de Toulon, Institute of Technology **2021 to 2023**
BSc in Electrical Engineering and Industrial Computing *Toulon, France*

Professional Experience

R&D Robotics Engineer **Mar. to Aug. 2026**
Capra Robotics - Internship *Aarhus, Denmark*

- Enabled fully automatic food-cart docking, with 18 successful dockings out of 18 in a 1-hour trial, by developing a LiDAR localization in the ROS 2 stack that runs on CPU in under 45 ms per cycle.
- Let the robot avoid obstacles its LiDAR misses by training a ground segmentation model (0.981 IoU) and integrating it into the C++ ROS 2 stack and Nav2 costmap on a Jetson Orin Nano.

R&D Innovation Engineer **Sept. 2023 to Mar. 2026**
Safran Data Systems - Apprenticeship *Les Ulis, France*

- Built a Python and C++ tool that lets flight-test engineers program embedded hardware in English, with a local LLM turning each request into editable blocks, which are then converted into C++ for ARM.
- Raised the success rate from about 35% to over 90% with deterministic code generation, then co-authored and presented the paper at the European Test and Telemetry Conference 2025.

Robotics Software Engineer **May to July 2023**
Blue Frog Robotics - Internship *Paris, France*

- Wrote the I2C driver in C for an 8×8 time-of-flight sensor in the head of the Buddy social robot, along with a gesture detector that triggers its behaviors.

Robotics Projects

LaMain **Since 2026**
Open-source dexterous hand for the SO-101 arm *github.com/arthur-mourgue/LaMain*

- Co-designed a fully 3D-printable, low-cost and lightweight 3-finger hand for the SO-101 arm, with 5 DoF, under 200 g and 200 €.
- Built a working prototype, now being fitted with fingertip tactile sensors to collect visuo-tactile demonstrations and train manipulation policies with touch.

LePotager **June 2026**
Most Creative Project award, LeRobot hackathon in Copenhagen *github.com/arthur-mourgue/LePotager*

- Built in 48 hours a fruit-sorting system on an SO-101 arm, with an ACT imitation learning policy that uses camera images and tactile feedback from an experimental piezoresistive sensor to judge ripeness.

Clubs and Competitions

RoboCup 2023 with Robot Club Toulonnais, Scientific Challenge winner and 3rd in Middle Size League
French Robotics Cup 2024 and 2025 with Clubelek, the robotics club of INSA Lyon

Technical Skills

Programming	C++, Python, C, Bash, MATLAB/Simulink
Robotics	ROS 2, Nav2, Zenoh, Costmaps, RANSAC, ICP, SLAM
Perception & ML	PyTorch, CUDA, ONNX, YOLO, SegFormer, ACT, VLA, JEPA, LLMs, AI
Systems & Embedded	Linux, Docker, Git, NVIDIA Jetson, Intel RealSense, OAK-D
Languages	French (native), English (C2, Cambridge Linguaskill)